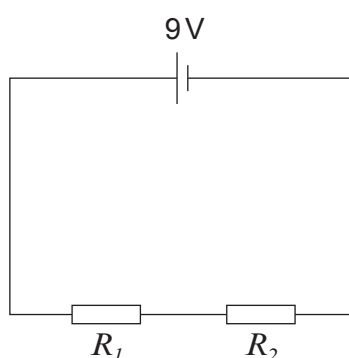


5. Jamie is investigating an electrical circuit which has two **identical** resistors, R_1 and R_2 . The circuit that he built is shown below.



- (a) Complete the sentences by underlining the correct word(s) in the brackets.

- (i) In this series circuit the current in R_1 is (**the same as / higher than / lower than**) the current in R_2 . [1]
- (ii) In this series circuit the voltage across R_1 is (**the same as / higher than / lower than**) the voltage across R_2 . [1]

- (b) The voltage across R_1 is 4.5 V.

- (i) Each resistor was found to have a resistance of $2.25\ \Omega$.

I. Use the equation:

$$\text{current} = \frac{\text{voltage}}{\text{resistance}}$$

to calculate the current in R_1 . [2]

current = A

- II. Use the equation:

$$\text{power} = \text{voltage} \times \text{current}$$

to calculate the power developed in resistor R_1 . [2]

power developed in R_1 = W

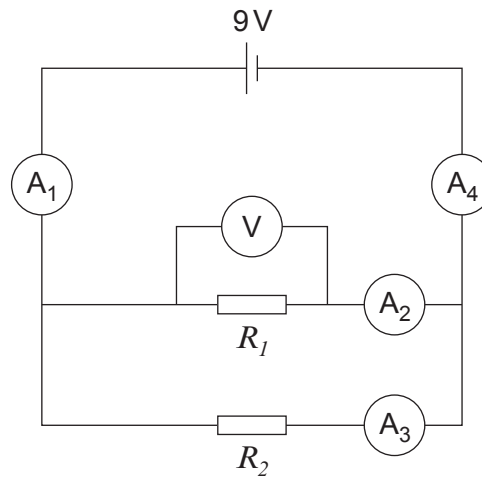


(ii) State the voltage across R_2 .

[1]

voltage across $R_2 = \dots\dots\dots$ V

(c) Jamie then built a parallel circuit using the same 2.25Ω resistors.



(i) State which ammeter (A_1 , A_2 , A_3 or A_4) measures the current through R_1 .

[1]

$\dots\dots\dots$

(ii) State the voltmeter reading across R_1 .

[1]

voltmeter reading = $\dots\dots\dots$ V

(iii) Ammeter A_2 reads 4 amps.

I. State the reading on ammeter A_3 .

[1]

ammeter reading = $\dots\dots\dots$ A

II. Calculate the reading shown on ammeter A_1 .

[1]

ammeter reading = $\dots\dots\dots$ A

